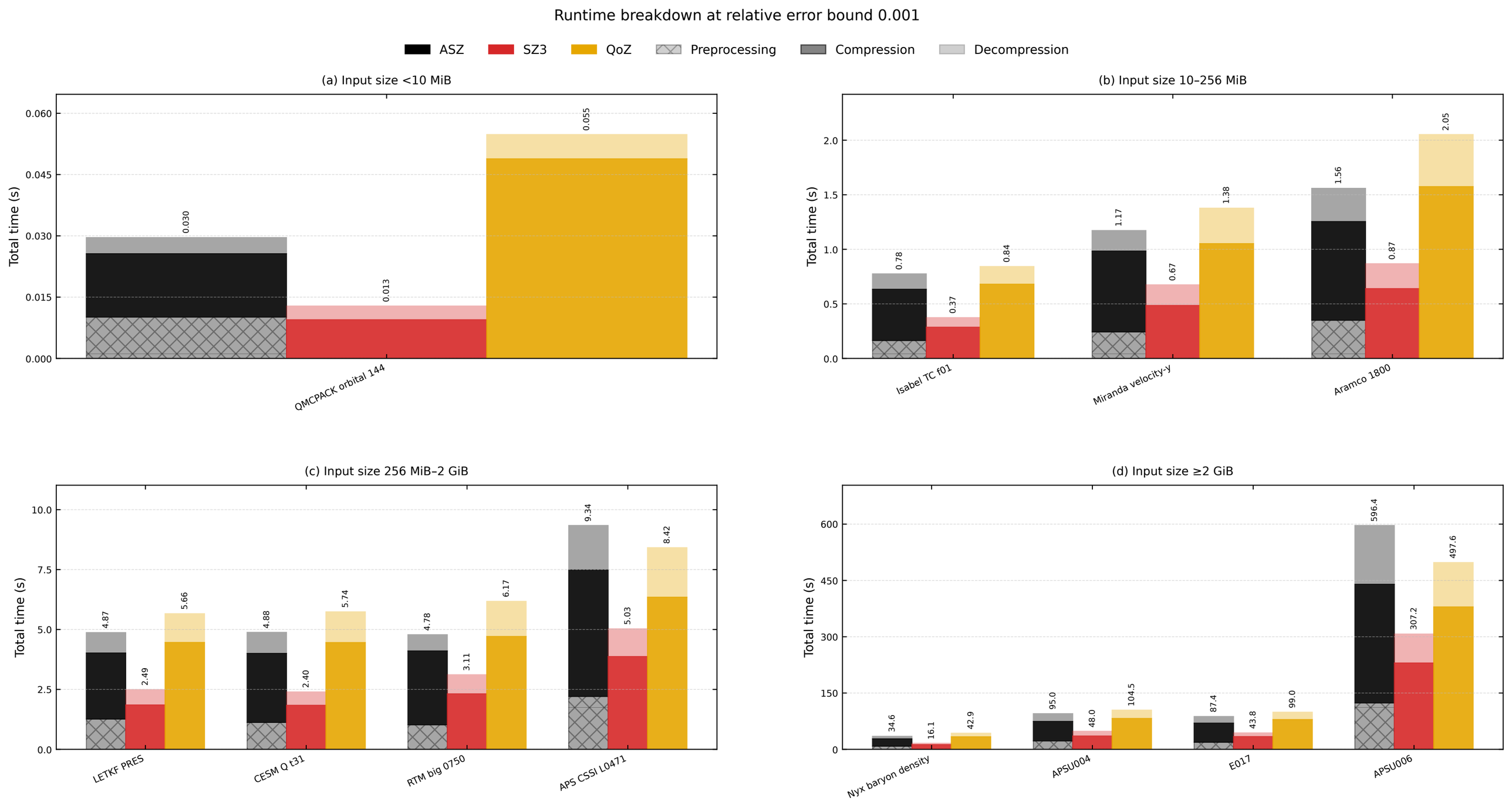
Runtime breakdown at relative error bound 0.001
ASZ
SZ3
QoZ
Preprocessing
Compression
Decompression
(a) Input size <10 MiB
Total time (s)
0.060
0.045
0.030
0.015
0.000
0.030
0.013
0.055
QMCPACK orbital 144
(b) Input size 10–256 MiB
Total time (s)
2.0
1.5
1.0
0.5
0.0
0.78
0.37
0.84
1.17
0.67
1.38
1.56
0.87
2.05
Isabel TC f01
Miranda velocity-y
Aramco 1800
(c) Input size 256 MiB–2 GiB
Total time (s)
10.0
7.5
5.0
2.5
0.0
4.87
2.49
5.66
4.88
2.40
5.74
4.78
3.11
6.17
9.34
5.03
8.42
LETKF PRES
CESM Q t31
RTM big 0750
APS CSSI L0471
(d) Input size ≥2 GiB
Total time (s)
600
450
300
150
0
34.6
16.1
42.9
95.0
48.0
104.5
87.4
43.8
99.0
596.4
307.2
497.6
Nyx baryon density
APSU004
E017
APSU006